CS/S-25/CS4T002/02 BT23002OCS

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

B. Tech Summer / ~~Winter~~ - 2025

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering **Sem:- IV**

Subject: Design & Analysis of Algorithm **Subject Code: CS4T002**

Marks: 100 **Date of Exam: 28/04/2025** **Time: 3 Hrs.**

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

CO1: Recall basic concepts of algorithm in analysis and Design of algorithms.

CO2: Examine Recurrence relations, solutions of recurrence of searching sorting methods

CO3: Apply Greedy methods used for analysis and Design of Algorithm

CO4: Analyse Dynamic Programming concepts in designing algorithm

CO5: Evaluate advanced techniques and tools available for algorithm analysis and development

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Enlist the characteristics of an algorithm.	L1/CO1	2
Q.2	State the need of analysing algorithm.	L1/CO1	2
Q.3	Classify the different algorithm Design Strategies.	L1/CO1	2
Q.4	Identify different types of complexities used to find efficiency of algorithm.	L2/CO1	2
Q.5	Write the different steps of Divide and Conquer Strategy.	L3/CO2	2
Q.6	Rewrite the concept of IPL and EPL with the relation between IPL and EPL.	L2/CO2	2
Q.7	State the principle of Job Sequencing with Deadline Problem.	L2/CO3	2
Q.8	Write difference between top-down and bottom-up dynamic programming approaches.	L3/CO4	2
Q.9	Define Multistage graph with example.	L1/CO4	2
Q.10	State the rules for solving N Queen problem.	L1/CO5	2
Q.11	Give the different graph traversal and tree traversal techniques.	L2/CO5	2

Q.12	Define Promising and Non-Promising Node.	L1/CO5	2

SECTION B

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Discuss different types of Asymptotic Notations.	L2/CO1	4
Q.14	Write an algorithm for Binary Search.	L3/CO2	4
Q.15	Draw split & merge tree for array of 10 elements.	L3/CO2	4
Q.16	Write short note on Spanning tree and Minimum Cost Spanning Tree.	L3/CO3	4
Q.17	Differentiate between Dynamic Programming and Divide & Conquer Strategy.	L4/CO4	4
Q.18	Sketch any two solutions for 4-Queen Problem and 8-Queen Problem.	L3/CO5	4

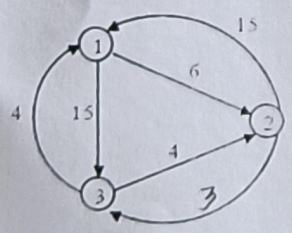
SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Solve the following recurrence $t_n=[\ 3t_{n-1} + 4t_{n-2},\ tn = 0 \text{ if } n=0,\ tn = 5 \text{ if } n=1]$	L3/CO1	7
Q.20	Find the average number of successful and unsuccessful search. Also find internal & external path length. Consider the array given below to perform binary search: a=[-15, -6, 0, 7, 9, 23, 54, 82, 101, 112, 125, 131, 142, 151] Also give the complexity of binary search.	L1/CO2	7
Q.21	Explain the Knapsack algorithm to find an optimal solution. Find optimal solution to the knapsack instance. No. of objects = 7, capacity = 15	L2/CO3	7

Objects	1	2	3	4	5	6	7
Weight	2	3	5	7	1	4	1
Profit	10	5	15	7	6	18	3

Q.22	Discuss an algorithm for Floyd Warshall and calculate distance and path matrix of following.	L5/CO4	7

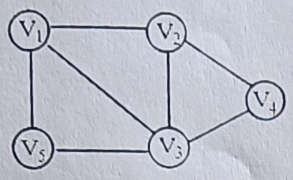


Q.23	Apply Longest Common Subsequence algorithm for following string & generate LCS.	L3/CO4	7

X = EXPONENTIAL

Y = POLINOMIAL

Q.24 Explain the Graph Coloring and Solution Space Tree. Apply Graph Coloring on the given graph. State the total number of possible solutions by using solution space tree. L5/CO5 7



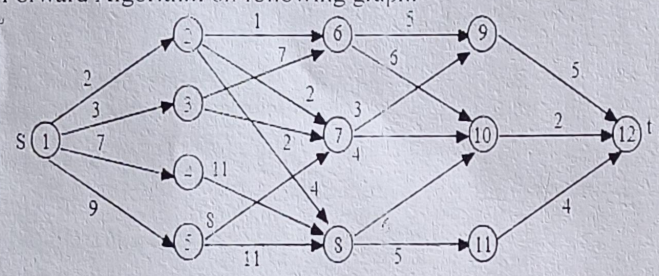
SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	**a)** Describe the Amortized Analysis & different methods used in Amortized analysis.	L6/CO1	
	b) Find the multiplication of the following matrices A & B given below using Stressen's matrix multiplication Algorithm.	L1/CO2	

$$A = \begin{bmatrix} 4 & 5 \\ 7 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 5 & 9 \end{bmatrix}$$

Q.26	Generate path from source to destination by applying Backward and Forward Algorithm on following graph.	L5/CO4	12



Q.27	Write Short notes on following:	L5/CO5	12

a) P Class Problem

b) NP – Hard

c) NP – Complete

d) Deterministic Algorithm